

Submitted via email to [REDACTED]

March 14, 2025

RE: J&J Response to National Science Foundation request for information on behalf of the Office of Science and Technology Policy, requesting input on the [Development of an Artificial Intelligence \(AI\) Action Plan](#).

Dear Faisal D'Souza and members of the National Coordination Office,

Thank you for the opportunity to provide comments on the Development of an Artificial Intelligence (AI) Action Plan ("Plan"). We commend the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation actions to consider input from varied stakeholders on the development of the Plan, as directed by a Presidential Executive Order on January 23, 2025, to define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation. We look forward to action on these crucial issues for patients, health care providers, and innovators across the healthcare ecosystem.

At Johnson & Johnson, our people are driven by a passion to achieve the best version of health for everyone, everywhere, for as long as possible. In the next decade, we will see more transformation in health than in the past century – and we are ready to lead the way. Focusing exclusively on transformational healthcare innovation allows us to move with purpose and speed to tackle the world's toughest health challenges. Innovating across the full spectrum of healthcare solutions puts us in a unique position today to deliver the breakthroughs of tomorrow.

Our strength in both biology and medical technology means we're accelerating advances in care – from cell therapy to AI-assisted robotic surgery. We are using our wide range of expertise to address healthcare challenges that can be tackled by both medical technology and innovative medicine such as cancer, cardiovascular disease, and eye health. Our reach and depth across a continuum of healthcare and technology solutions give J&J the ability to profoundly impact health for humanity.

How We are Using AI

At J&J, we are applying AI across our business, focusing on finding solutions to big questions and advancing our impact on patients. We believe AI, when used at scale, has the potential to accelerate our ability to advance human health, including around chronic diseases such as autoimmune and mental health disorders. AI allows us to analyze larger-than-ever data sets in ways never before possible to find patterns and trends that are helping to transform our innovation engine.

Through AI and data science, we can better understand the drivers of diseases, transform how we innovate to create treatments, design more personalized healthcare for patients, and deliver efficiency and precision for clinicians and surgeons to improve medical outcomes. We are using AI, leveraging de-identified data, in a multitude of ways across our sectors:

Accelerating drug discovery: AI helps researchers to understand diseases better, find new targets, and design potential medicines faster. It can also speed up the processes of improving molecules or potential medicines around features that are hard to test in clinical trials, such as toxicity and immunogenicity.

Increasing access to clinical trials: We are using machine learning (ML) algorithms to analyze real-world health data from resources like de-identified electronic medical records, labs, claims and imaging to identify and prioritize geographies and healthcare organizations where patients are being treated.

Driving precision medicine: We are applying to AI and ML to data sets – from digital images to sound recordings – to identify novel biomarkers and clinical endpoints that can help us develop and deliver the right medicines for the right patients at the right time – across oncology, immunology and neuroscience.

Predicting demand more accurately: Our digitally integrated system Advance Case Management uses these ML algorithms to predict demand more accurately for knee, hip and other joint surgical implants, to ensure these products are stocked when and where they're needed.

Analyzing health records with AI: We are deploying an AI-derived eye disease risk alert system to eye care offices, using electronic health records from more than 80 million patients to improve patient care and encourage referral of at-risk patients.

Training surgeons with insights: In the operating room, surgical video is frequently taken during procedures to provide education, research, and quality improvement to medical professionals. We are training AI algorithms to create a “highlights reel” of these videos in a matter of minutes, so surgeons can re-watch significant events from their procedures. The goal is to use these enriched surgical videos to run algorithms to understand the behaviors, tactics and movements that create positive and negative outcomes during surgery.

Efficacy, Accuracy, and Transparency: Healthcare Considerations

While it is undeniable that AI for healthcare offers numerous opportunities to improve patient care, there are significant opportunities to ensure that AI can be developed and deployed in a way that fosters trust and appropriate use across healthcare systems.

Access to quality data: As data fuels AI, we need to access large enough data sets to train models. This requires rational approaches to determining data ownership, control, and governance structures. We must also consider the quality of the data sets. If data used to train the AI model has limitations and does not adequately represent the target population, one could have outputs that are not representative enough.

Safeguarding Privacy: Another key element for safe, ethical, and transparent AI use is privacy. We operate in a highly regulated industry where we use a variety of types of data, including administrative and claims data, clinical data, genomic data, patient-generated data, surgical video, and social determinants of health data that move through a variety of transmission networks. This data is essential to continued innovation, discovery, evaluation, and speed for the delivery of healthcare products and services to patients and consumers.

We support adoption of a comprehensive national privacy law and associated standards to help ensure a more consumer- and patient- friendly approach to managing personal information while ensuring consistent privacy protections, allowing a greater flow of data, and maintaining adequate protections. A federal law would help reduce variability across multiple governments and government agencies, consequently reducing regulatory costs and increasing competitiveness to attract investments.

Reskilling and Upskilling for Workforces: As technologies like AI take hold, the skillsets needed to excel in healthcare are shifting. Healthcare professionals need to have a thorough understanding of digital technologies for healthcare systems to effectively guide patients.

- We need investment in workforce skilling via lifelong learning programs and university education, equipping the workforce with the reskilling and continued learning opportunities required to embrace ongoing technological developments to maximize the positive impact of AI. We also need more life science education and training programs to improve the application and customization of technologies like AI to healthcare needs.
- Inclusive approaches must be considered in all aspects of AI, to include selecting the issues to address/problems to solve using AI, and training and hiring a workforce with varied backgrounds to serve as data scientists, programmers, attorneys, program managers, etc. We believe that fostering participation across broad populations will help enable data generation that simultaneously improves the authenticity of data sets and the inclusivity of data-driven insights.
- We advocate for updates to curriculum from elementary through secondary schools to account for new content to foster understanding of the technology and the soft skills required to appropriately and responsibly use these new technologies and tools.

Ethical and Regulatory Considerations

Ethics: We believe ethical considerations should be at the forefront of how we are applying AI models, and that regulation has a key role to promote the responsible and transparent usage of AI. Well-established practices and standards exist for demonstrating clinical validity and utility and connecting these efforts to meaningful outcomes for patients. It is important to start by assessing the existing research practices used for creation of treatment guidelines and approvals of new therapies to assess any AI-related gaps.

Rational Regulation: Recognizing the accelerated speed of AI continuous development, it is crucial to promote consensus-based standardization to assure a thriving and innovative ecosystem, avoiding a fragmented system that could hinder investment in innovation in the long term. We support federal law and regulations to foster consistency and predictability. Any policy regarding AI should have a clear scope and framework, facilitating implementation and avoiding complexity, to build trust between citizens, developers, deployers and users and create a favorable environment that fosters medical innovation.

- Regulatory efforts should focus on a risk-based approach, a transparent framework for compliance and enforcement and versatility of regulation to promote innovation with the aim to pursue fair, responsible, secure, and transparent use of AI. Effective implementation of any regulatory approaches can be enhanced through thoughtful public-private partnerships and creation of 'regulatory sandboxes' to enable sharing of applications and foster cross-learning and collaboration to solve complex issues or problems.
- Policy development should begin with an assessment of existing policies and regulations, including applicable sector-specific laws and guidelines, such as the Health Insurance Portability and Accountability Act of 1996 (HIPAA). Highly regulated sectors, like the healthcare industry, have existing frameworks for addressing relevant risks (e.g., safety, security, privacy), and these frameworks should be evaluated for their contribution and applicability to AI. We also support leveraging internationally recognized standards such as the International Organization for Standardization (ISO).
- Overly strict AI regulations could stifle innovation in healthcare, hindering the development and implementation of advanced AI-driven solutions, slowing down progress in medical research and patient care. The risk increases when taking into consideration the present and potential future lack of consistency across regulatory agencies (within a market at the federal and state levels and across markets).
- Federal laws and regulations should provide liability protections when adherence to regulatory requirements, such as those established by FDA, is demonstrated. This will encourage use of high-quality regulatory practices and advancement of trustworthy AI applications.
- Misalignment across laws in different countries and/or states, leading to conflicting laws and definitions, is a risk for long term investment in digital healthcare technology. Achieving harmonization of AI regulations to facilitate the sharing of knowledge and technologies across borders is a complex but key challenge.

Access to AI-enabled technologies. Digital health technology is reshaping healthcare systems around the world, and innovation continues to move at a rapid pace. Digital health technologies -- tools, like software and mobile apps, are increasingly AI enabled and have potential to improve care delivery, quality, and patient outcomes; however, predictable coverage and payment for these tools is not always obvious or available.

As these technologies advance and come into more frequent use, changes may be needed to ensure predictable, equitable and sustainable coverage and payment. We support coverage and reimbursement approaches that facilitate the integration of digital health technologies to promote better health outcomes, increase value or improve the patient experience.

Intellectual Property. Different AI elements could be subject to intellectual property (IP) protection, including training data, AI models, model improvements, and model outputs. However, lack of clarity and differences in existing IP laws present obstacles. To enhance innovation with AI, we need consistent protection of existing IP rights in the use of AI through clear and consistent application of IP legal frameworks and clarity regarding liability. In addition, we call for clarity regarding liability and accountability implications related to IP ownership of AI models in the healthcare context.

Conclusion

We recognize the power and promise of AI in healthcare and believe this is a collective effort. We value our role as collaborator and innovator in healthcare, contributing new ideas, solutions, technology, partnerships, and perspectives on AI policy. We are focused on increasing engagement and cocreation with patients, providers, and policymakers to raise the understanding on how harnessing the potential of AI in healthcare can help everyone.

J&J looks forward to serving as a resource and providing additional thoughts about the AI Action Plan to assure AI is ethically, safely, and efficiently integrated into our society. Thank you for your attention to this critically important issue. If you have any questions or we can provide any assistance on issues related to healthcare, please e-mail Leif Brierley at [REDACTED]

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Sincerely,

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